

3D DENTAL PRINTING COURSE — DENTIST EDITION

# Complete Guide to 3D Printing for Dentists

*The quick visual guide: what to orient, how to arrange parts, which supports to use — for models, bite splints, surgical guides and restorations.*

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5 simple rules, valid for (almost) every job

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SECTION 01

# 01

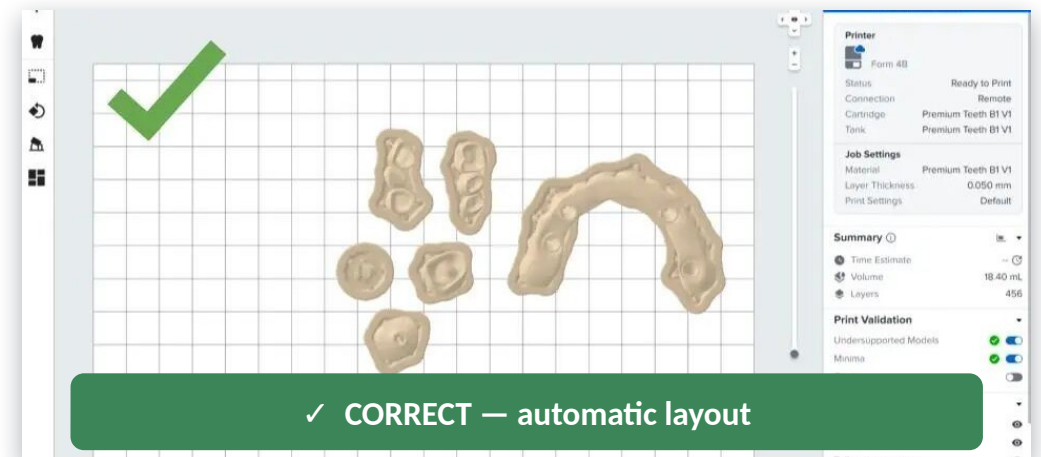
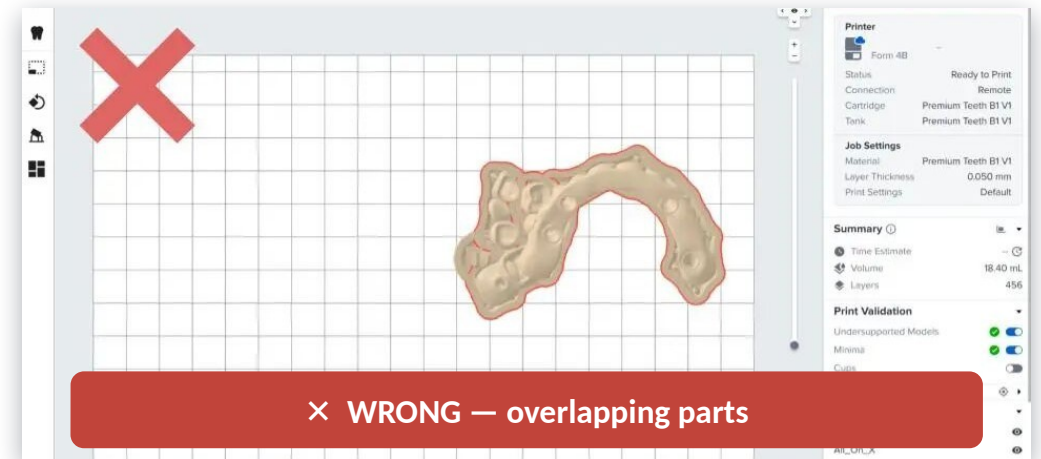
## The Golden Rules of Nesting

*Five simple rules, valid for (almost) every job you print in-office.*



# The 5 Golden Rules of Nesting

- ✓ Keep the intaglio surface (the one that will contact the patient's dentition or tissue) facing away from the build platform
- ✓ Models directly on the platform; for anatomical surfaces always use full bases and supports, for secure adhesion
- ✓ Generate supports automatically, then manually refine only the critical areas
- ✓ Never overlap parts: always check the green printability icon before starting the print
- ✓ Align the insertion axis (abutment, implant) with the platform's vertical axis





SECTION 02

# 02

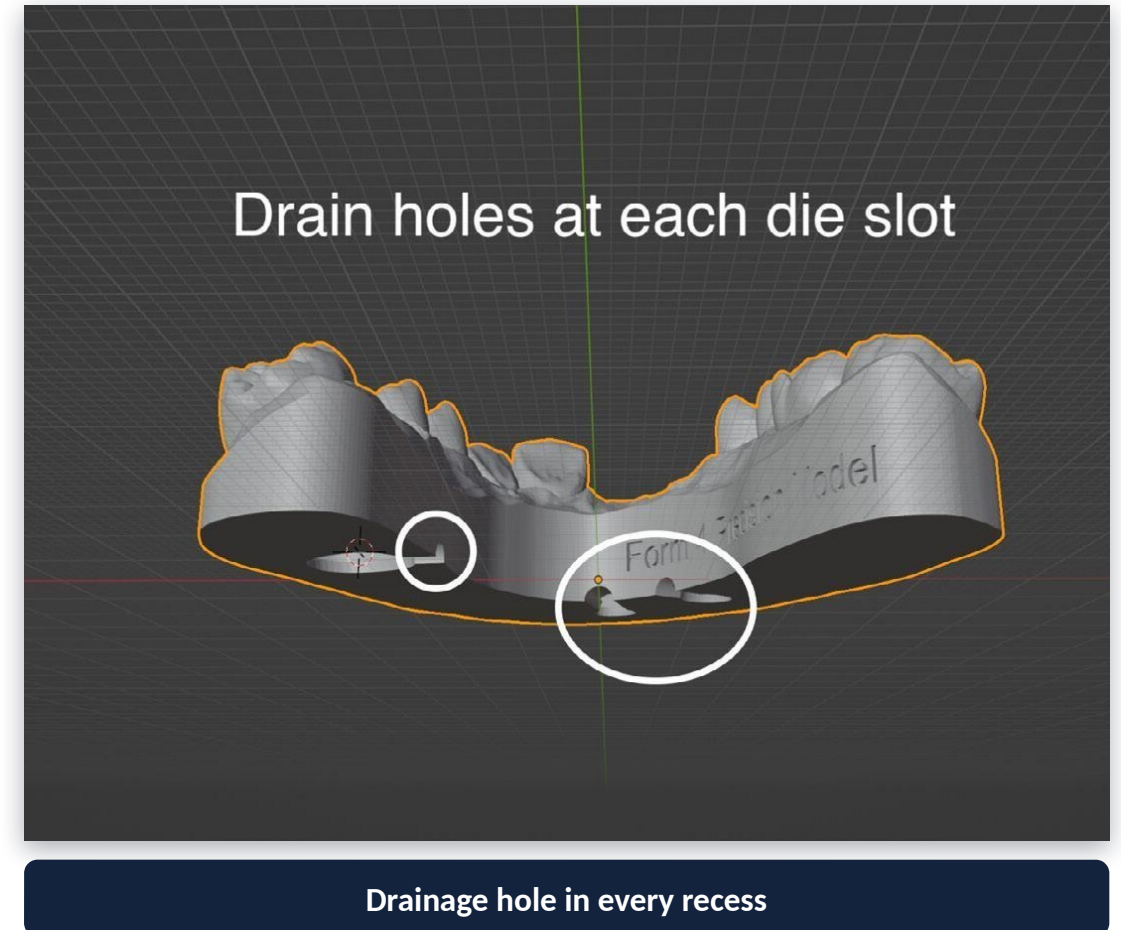
## Precision Models

*Orientation, drainage holes and tray layout.*



# How to orient them on the platform

- ✓ Print arches horizontally, directly on the platform: maximum precision, no base needed
- ✓ Add a drainage hole in every abutment recess: prevents resin from getting trapped (the "suction cup effect")
- ✓ For implant analog housings, holes are strongly recommended: they also let you visually check positioning
- ✓ 50  $\mu$ m layers for maximum clinical precision

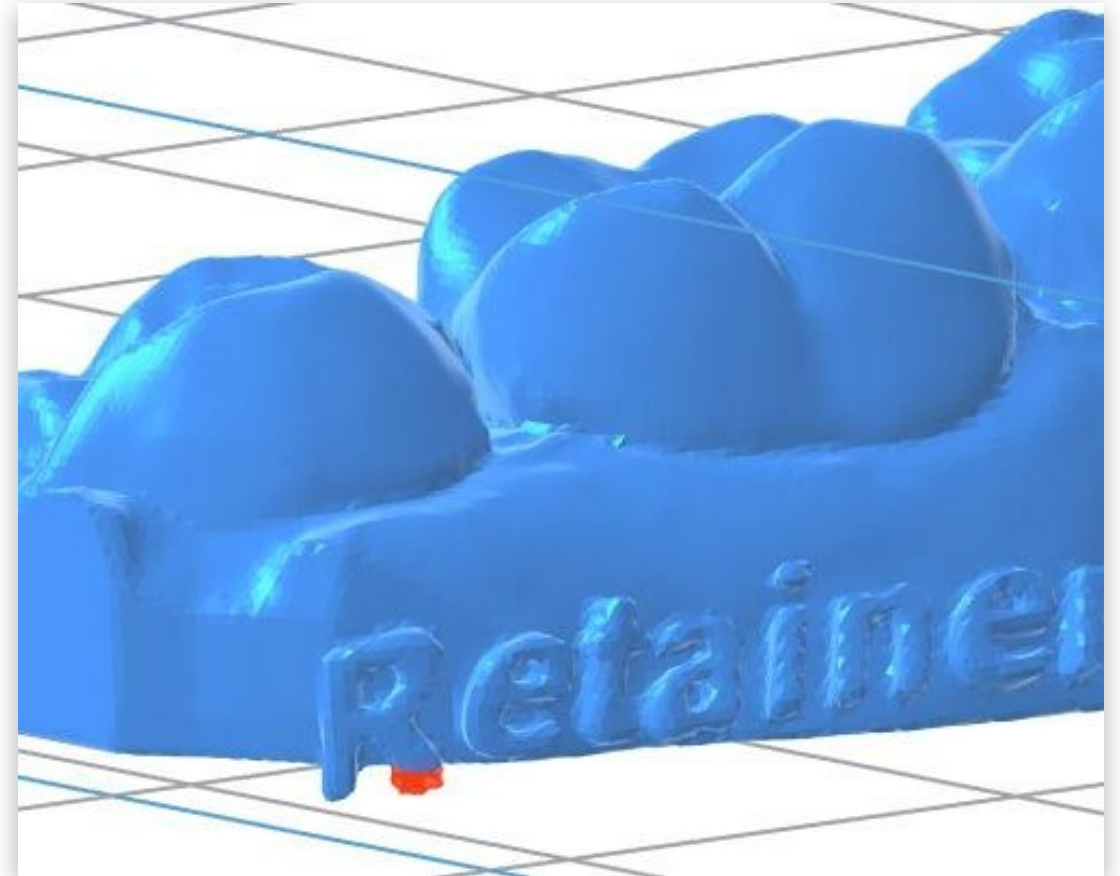


# Insufficiently supported surfaces or overhangs

- ! Diagonally developing surfaces, raised labels, clasps or brackets: if the base doesn't rest fully on the platform, the part isn't sufficiently supported
- ! Typical example: a raised printed label (e.g. the patient's name) protrudes from the base and prevents it from resting fully on the platform
- ! If not printed correctly, the protruding part can bend or detach during printing

## How to spot it before printing

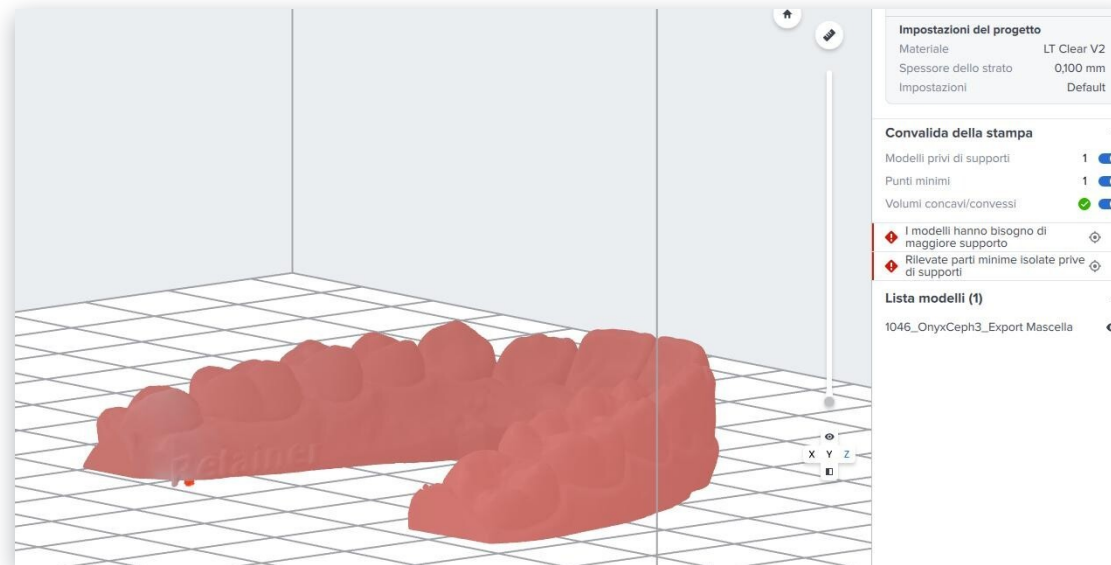
Without Print Validation checks enabled, this hazard isn't visible. Always turn them on: the software automatically flags unsupported points.



The label protrudes: the base doesn't rest flat

# Print Validation: identifies unsupported points

- ✓ The Print Validation panel automatically flags unsupported models and small isolated parts
- ✓ Always turn it on before printing: it identifies risks invisible to the naked eye

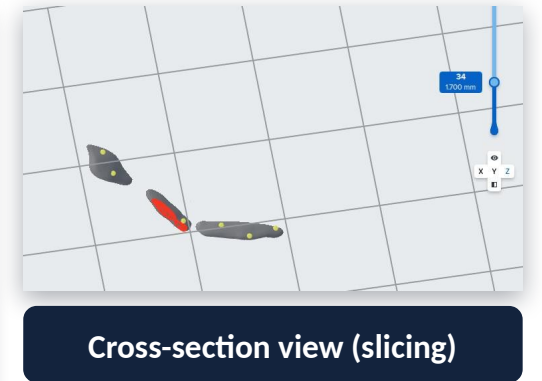
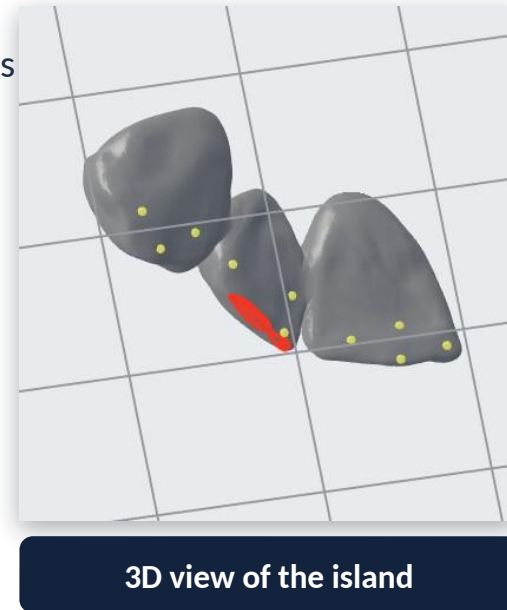


Print Validation: insufficient support warning



# Check layers with the side slider (slicing)

- ✓ The vertical slider beside the platform scrolls through the print layers: it shows exactly the cross-section at every height, layer by layer
- ✓ It's the most reliable tool for spotting separate islands that aren't sufficiently supported: the software highlights them in red at the exact layer where they appear
- ✓ Also use it to check that the model's base adheres fully to the platform in the first layers, with no gaps or uncovered spots
- ✓ Always check it before starting the print, alongside Print Validation



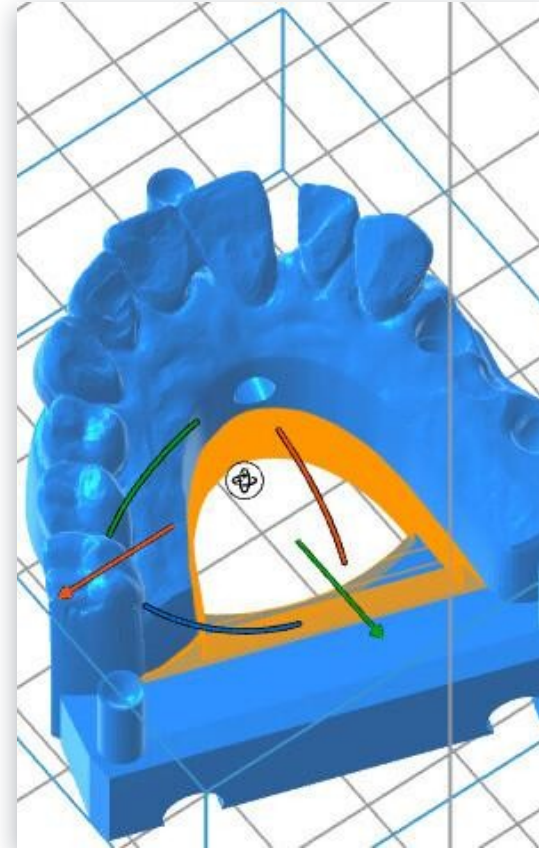
In both views, an island not connected to the rest of the model is highlighted in red: without a dedicated support it will detach during printing.

# The suction cup effect: concavities that trap resin

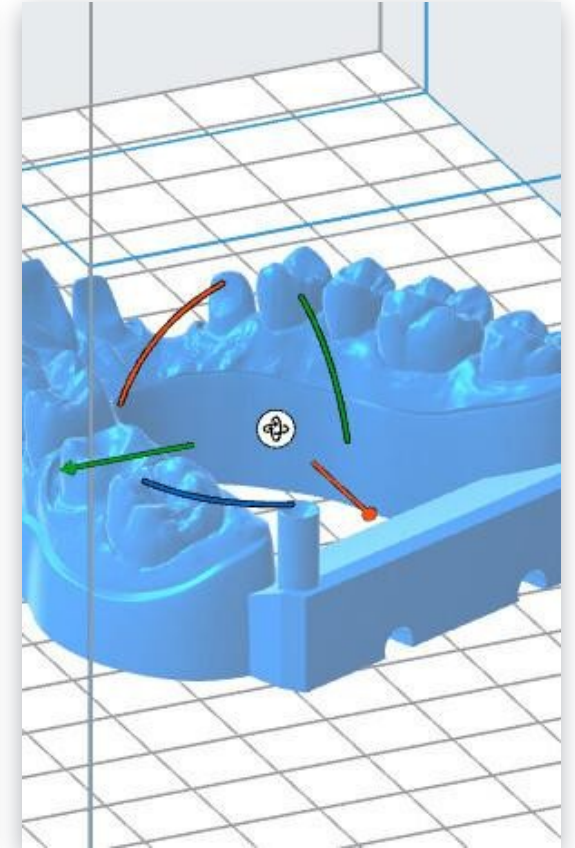
- ❗ Cavities like the palate or the inside of articulator bases: without a sufficient drainage hole, resin stays trapped inside (the "suction cup effect")
- ❗ The hole must pass through the ENTIRE base, articulator included — only then can trapped resin escape during printing and washing
- ❗ Always check the hole's path in transparency view before printing

## How to spot it before printing

Always turn on Print Validation: it automatically identifies concave volumes at risk of trapping resin.



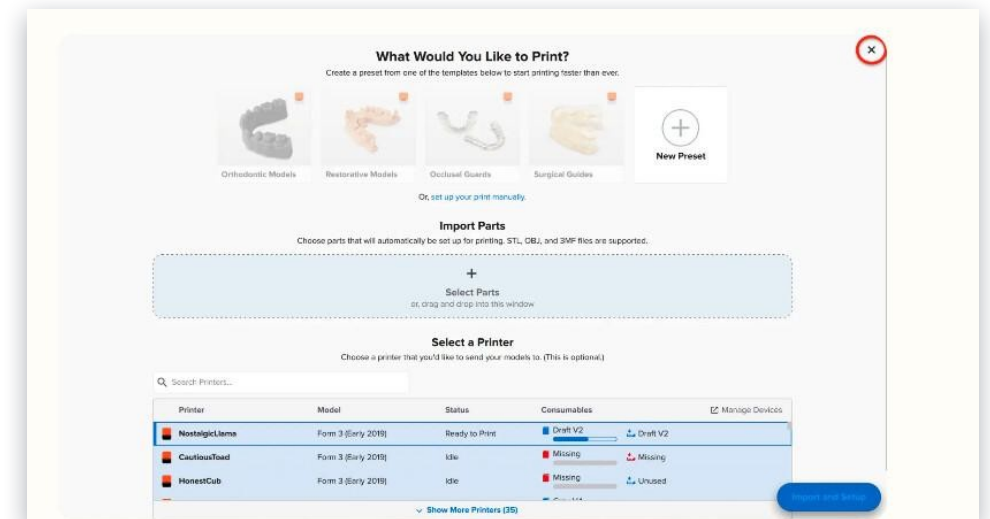
Incorrect side hole: trapped resin



Vent holes: clear interior area

# PreForm presets: automatic orientation

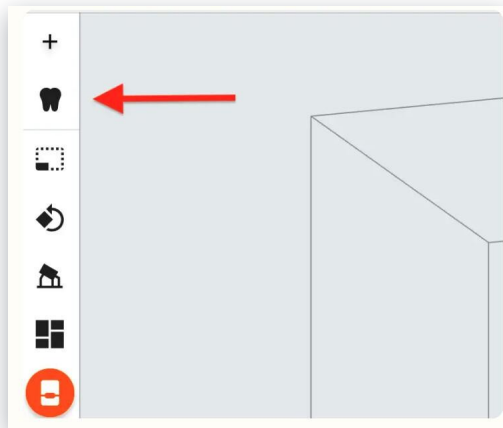
- ✓ On the job setup screen, select the preset best suited to the print type: models, bite/splints or surgical guides
- ✓ The software automatically orients the model according to best practices for the chosen category — for models, for example, it positions the occlusal surface away from the platform
- ✓ Presets are a starting point: always check the proposed orientation before printing — they don't replace visual inspection



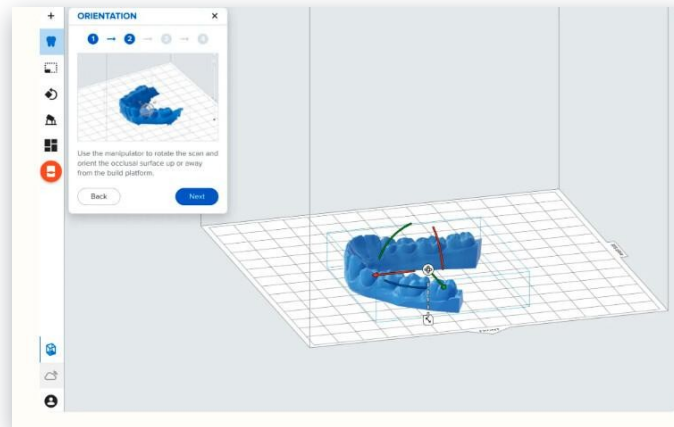
Choosing the preset in PreForm

# Scan to Model (1/2): find the tool and orient the scan

- ✓ In PreForm's Dental Workspace, open the Scan to Model tool: it's the tooth-shaped icon in the left-hand toolbar
- ✓ The software automatically orients the scan according to clinical best practices, ready for the next step



Scan to Model icon in the toolbar



Automatic scan orientation

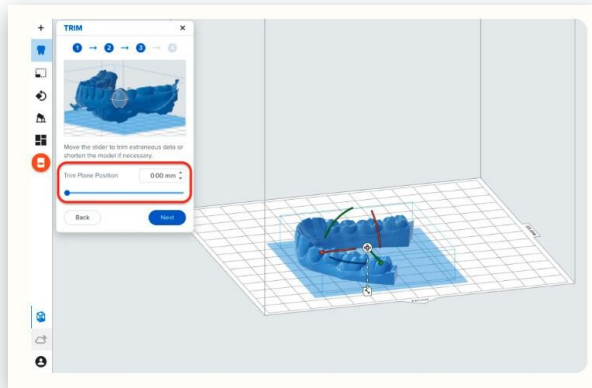
## Why it matters

Automatic orientation applies, in seconds, the same rules you'd follow manually, cutting file preparation time.

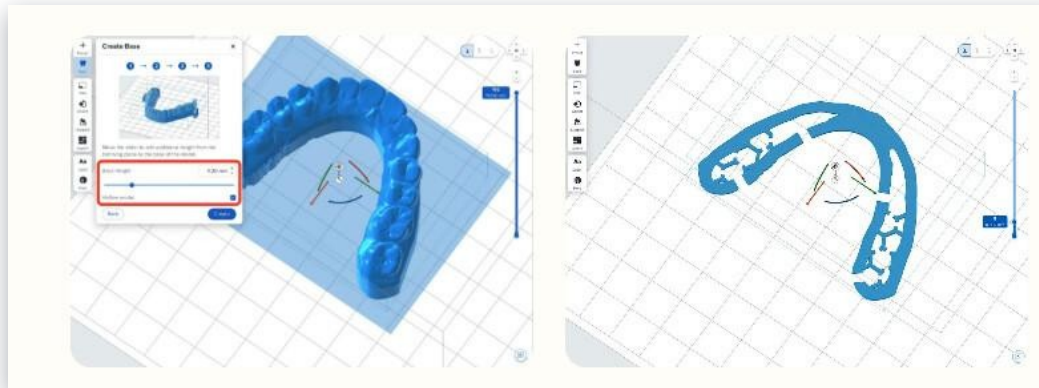
Source: Formlabs — "Preparing Dental Scans for Printing with Scan to Model" ([formlabs.com/support](https://formlabs.com/support))

# Scan to Model (2/2): refine and create the base

- ✓ Apply the automatic trim plane to clean up the scan from excess data: adjustable with the dedicated trim-plane position slider
- ✓ Add the base height with the dedicated slider: base ready in under 30 seconds
- ✓ Optional: enable "Hollow the model" to automatically generate the internal cavity and drainage holes



Trim plane adjustment



Base height and hollowed model

Source: Formlabs — "Preparing Dental Scans for Printing with Scan to Model" ([formlabs.com/support](https://formlabs.com/support))



# From print to finished model

1

## Remove

Detach parts and/or supports from the build platform

2

## Wash

Remove every trace of uncured resin

3

## Dry

Blow with compressed air for a few seconds, then leave the part on the bench for 30 minutes: the solvent must evaporate completely, or it will re-expand in the curing unit under heat

4

## Cure

Always follow the resin manufacturer's protocol

# Watch out for non-IPA solvents

- ! Non-IPA solvents (non-flammable and non-evaporating) do not evaporate from the part's surface
- ! They require a second wash under running water: otherwise they leave an oily residue on the surface
- ! They are not suitable for biocompatible materials: residue can compromise their clinical safety

## What to use instead

Always use a good-quality isopropyl alcohol (IPA), 99.9% — for example GIP103 — and it will evaporate completely without leaving residue.

# Curing: a critical step, not optional

## What curing actually does

It achieves the final cross-linking of the polymer chain and ensures the mechanical properties of the part. Only curing carried out according to the parameters (time, temperature, wavelength) specified by the resin manufacturer achieves both results.

- ! Every resin has a specific curing protocol: time, temperature and wavelength are not interchangeable between materials or devices
- ! Incomplete curing leaves residual monomer: the part stays brittle, deforms over time, and may not be biocompatible
- ! Don't rely on visual estimates: always use the curing unit and protocol specified by the manufacturer of the resin in use

SECTION 03

# 03

## Occlusal Bite Splints

*From scan to finish, step by step.*



## From scan to design

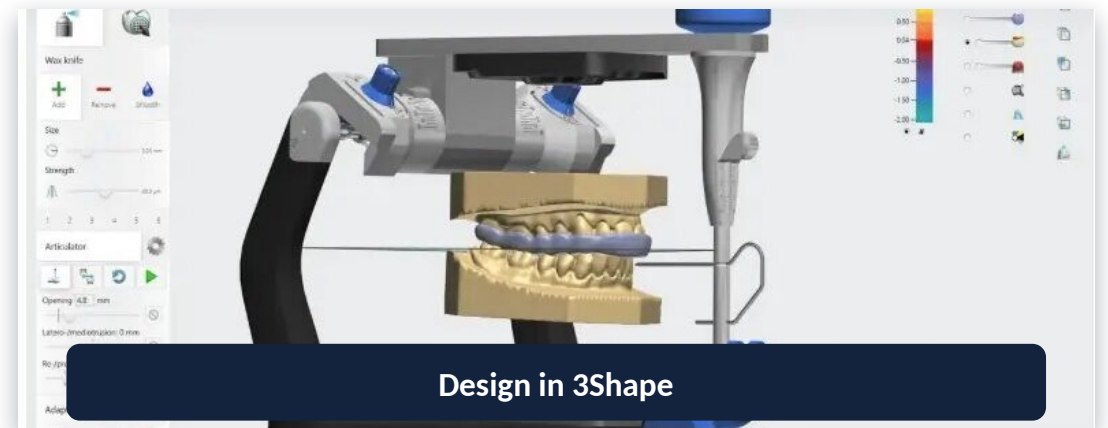
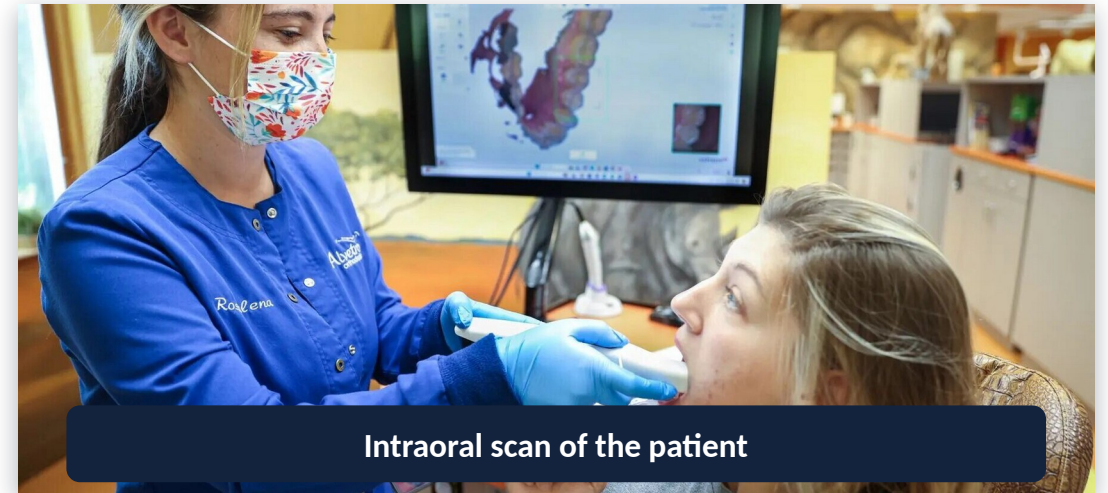
- ✓ Physical impression or intraoral scan of the patient
- ✓ Prescription with the desired bite type: rigid or soft
- ✓ Design in 3Shape or exocad, with the offset recommended for the chosen material and minimum thicknesses of at least 2 mm in the occlusal area and 1 mm in the side walls

#### Dental LT Clear Resin

Rigid, fracture-resistant — rigid occlusal bite splints. Offset 0.05–0.10 mm

#### Dental LT Comfort Resin

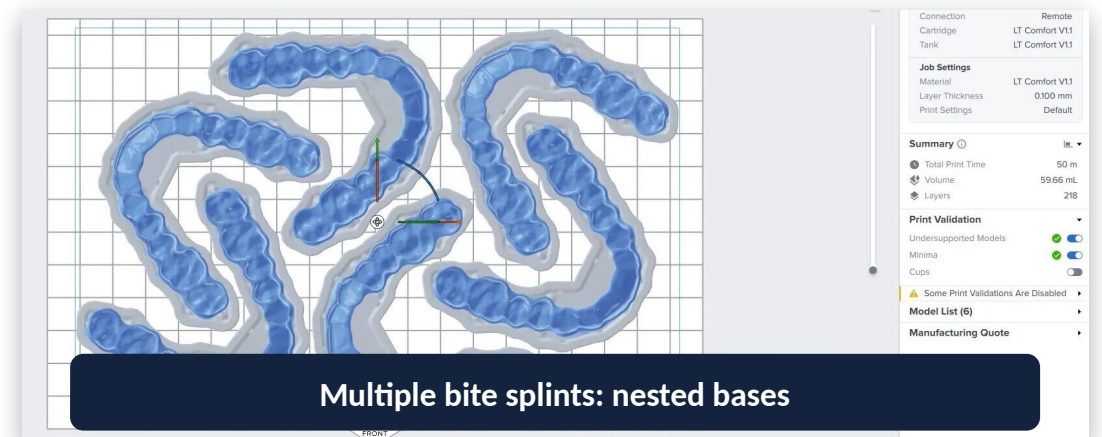
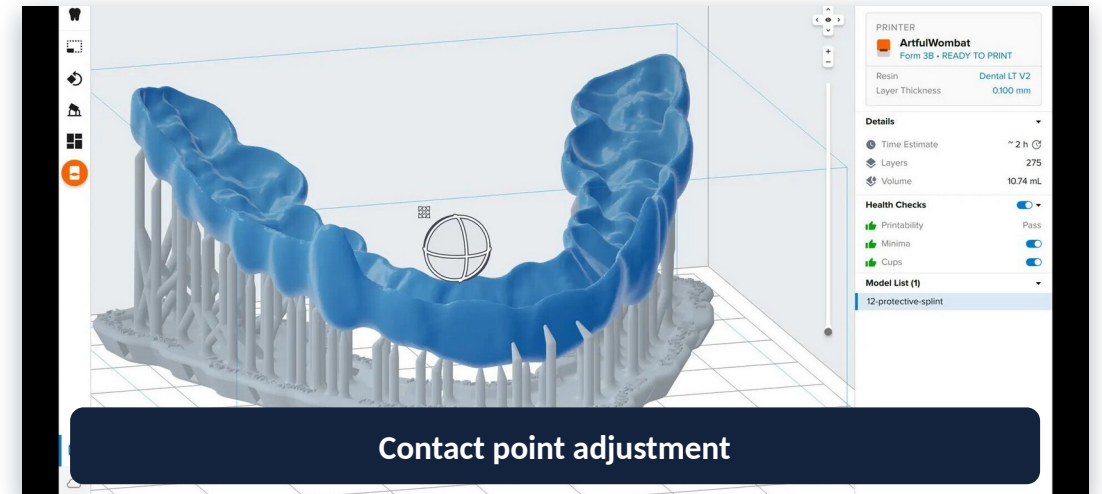
Flexible and comfortable — night guards, mouthguards. Offset 0.01 – -0.03 mm



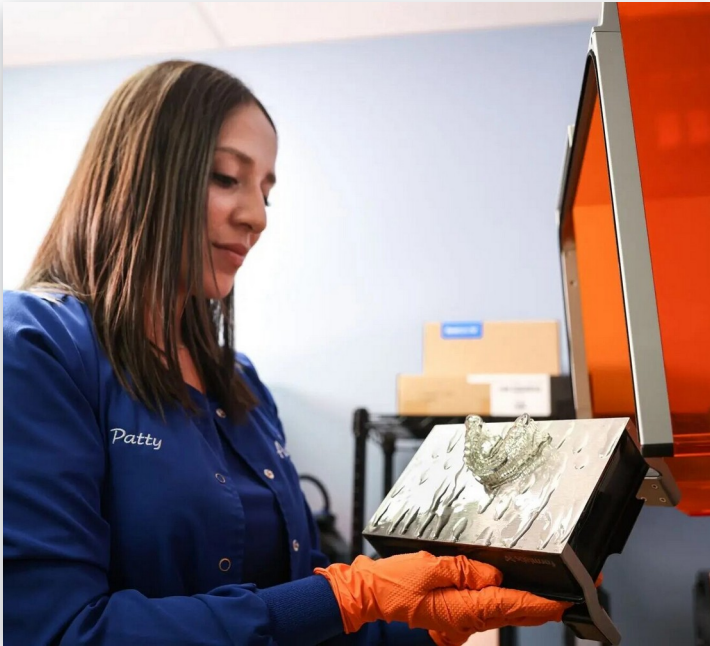


# Simple nesting in PreForm

- ✓ The Dentistry workspace automatically orients and supports the bite splint
- ✓ Always check the green printability icon before printing

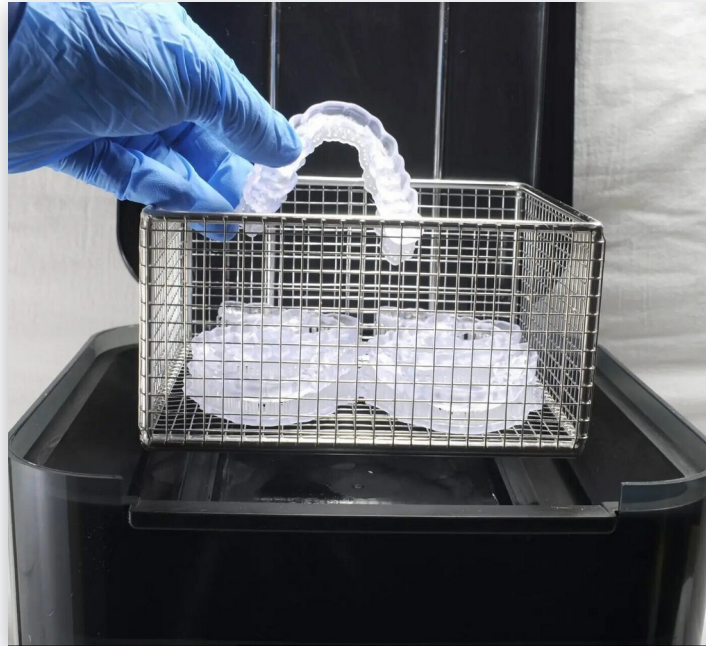


## From printer to patient (1/2)



### 1 • Print

Dedicated cartridge, platform and resin tank for bite splints



### 2 • Wash

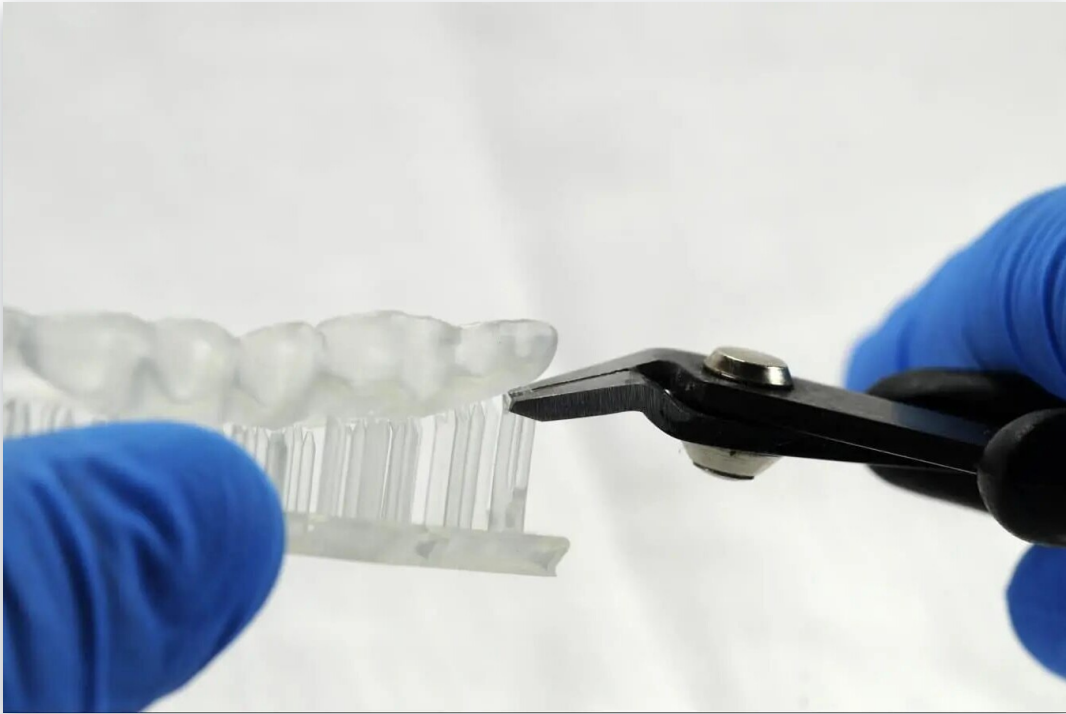
99% isopropyl alcohol (IPA), parts fully immersed



### 3 • Cure

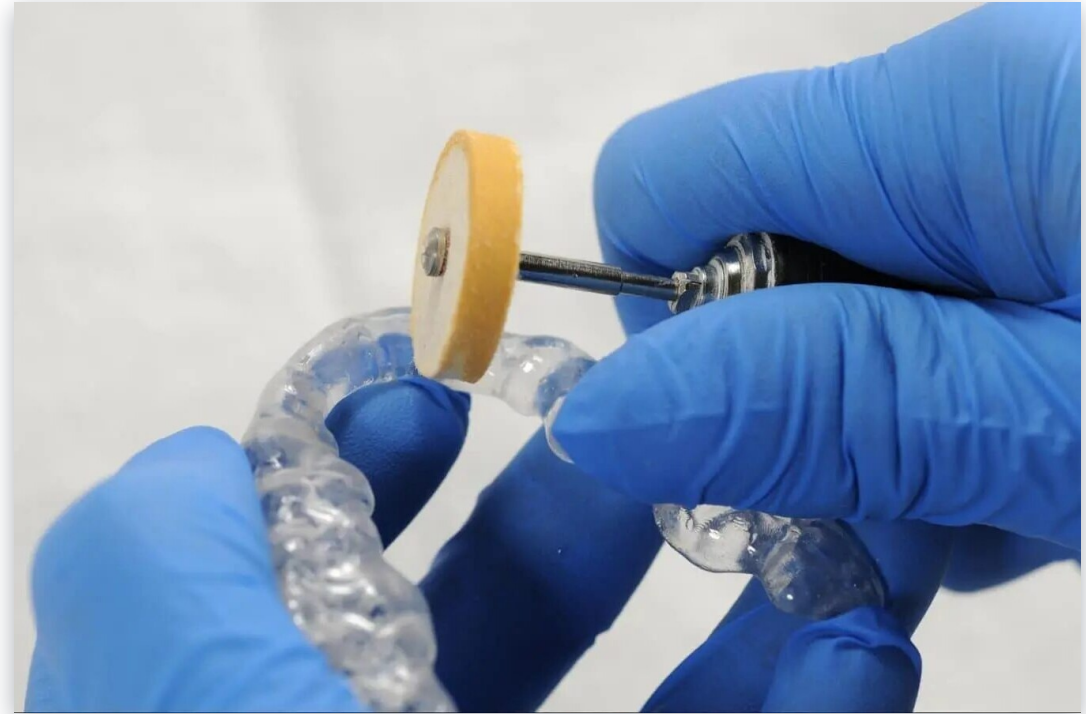
Form Cure, times specified by the material

## From printer to patient (2/2)



### 4 • Remove supports

Cut one at a time, don't tear

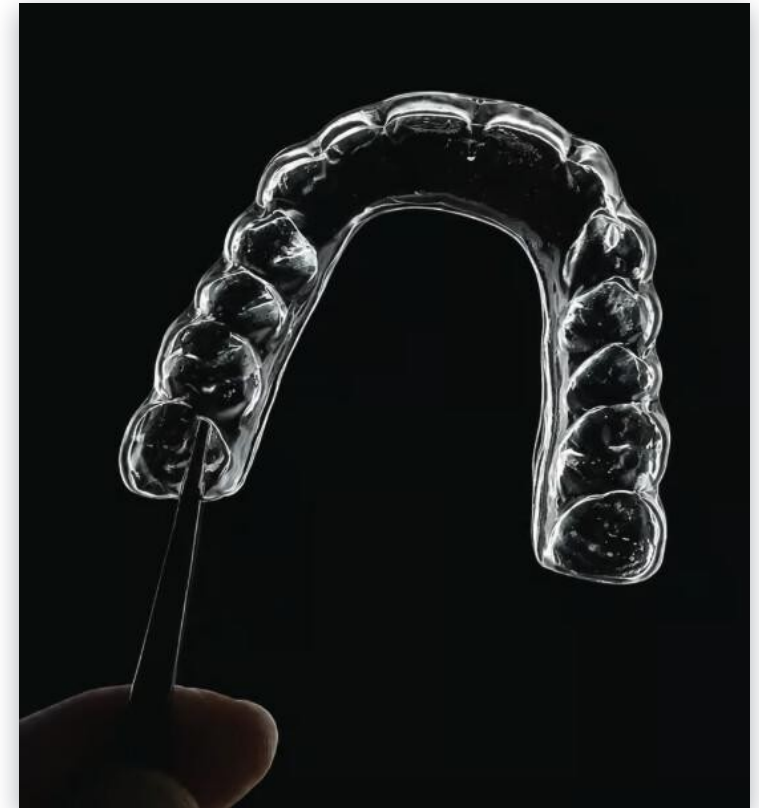


### 5 • Polish

Pumice and universal polishing paste for a mirror finish

## For those printing with Asiga Ultra: UltraGLOSS / LIFT resin trays

- ✓ UltraGLOSS / LIFT resin trays auto-polish bite splints and aligners directly during printing
- ✓ The result is a smooth, clear surface, ready for the patient without manual polishing
- ✓ 100/100 bite splints printed and accepted — zero failures, zero detachments



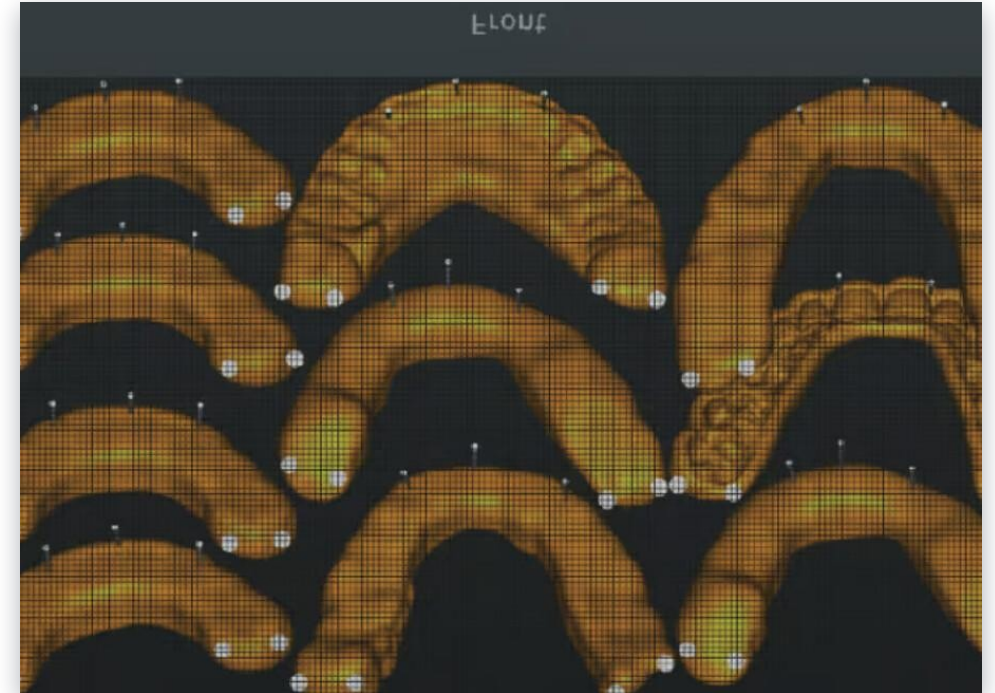
Bite splint auto-polished by the UltraGLOSS tray

Source: Asiga — “UltraGLOSS / LIFT” ([asiga.com/blog](https://asiga.com/blog))



## UltraGLOSS / LIFT Resin Trays: Orientation in Composer

- ✓ Near-vertical orientation (75–85° relative to the platform): reduces the contact cross-section per layer and improves resin drainage
- ✓ Sprues: 2 supports of 3 mm placed on the sides, at the distal ends of the arch — they detach without clinically relevant marks on the finished bite splint
- ✓ Central supports: 2–3 flexible 1 mm supports in the anterior area, positioned as cantilevers and parallel to the platform, to stabilize the incisor area during printing



Orientation in Asiga Composer

Source: Asiga — “UltraGLOSS / LIFT” ([asiga.com/blog](https://asiga.com/blog))

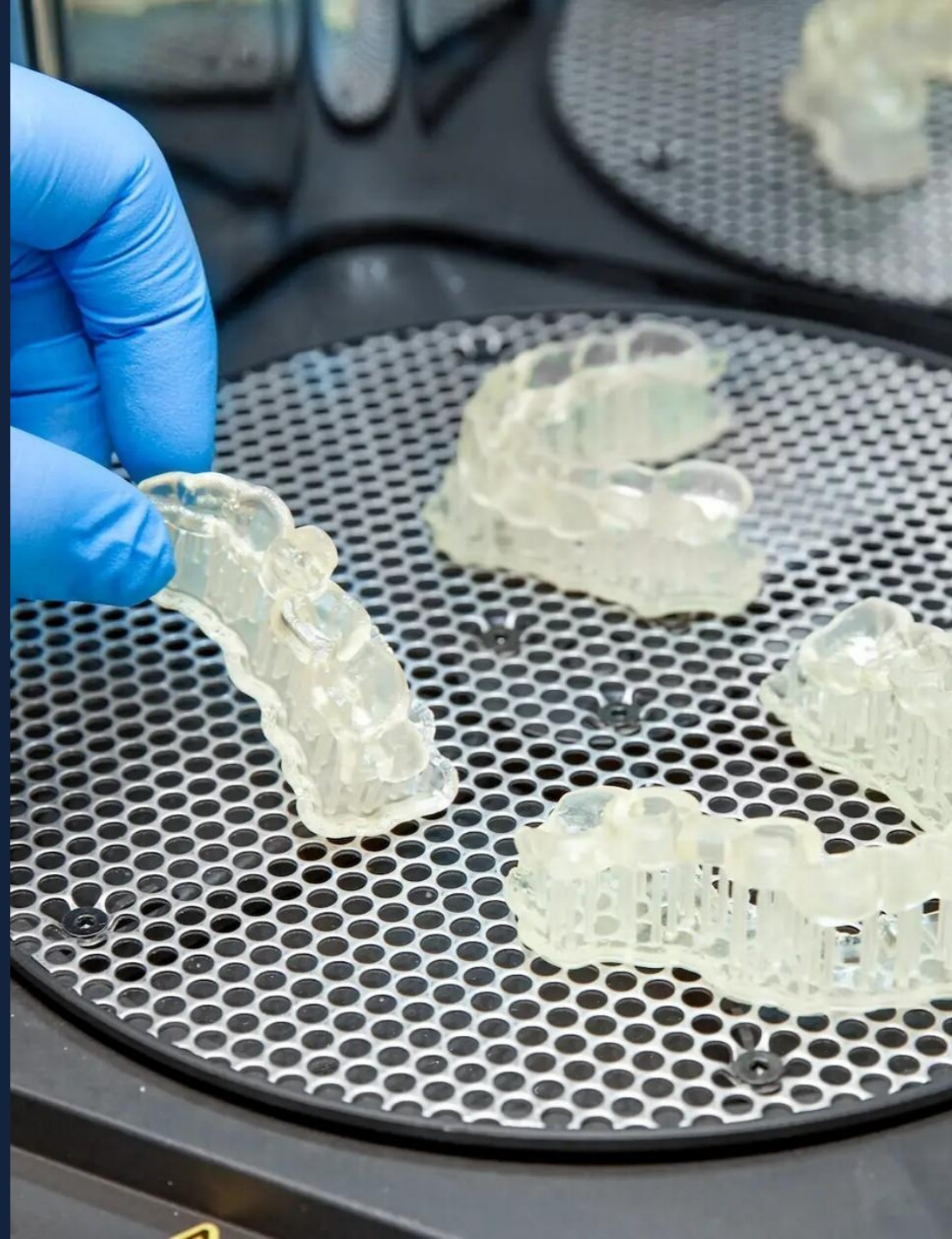


SECTION 04

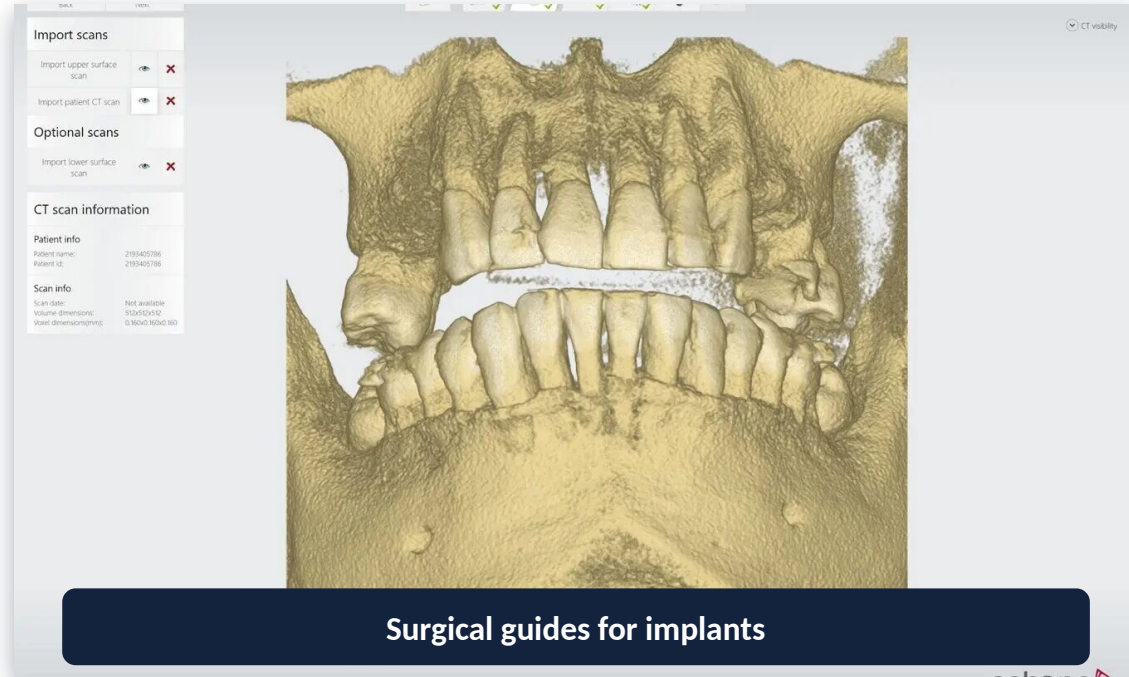
# 04

## Surgical Guides and Templates

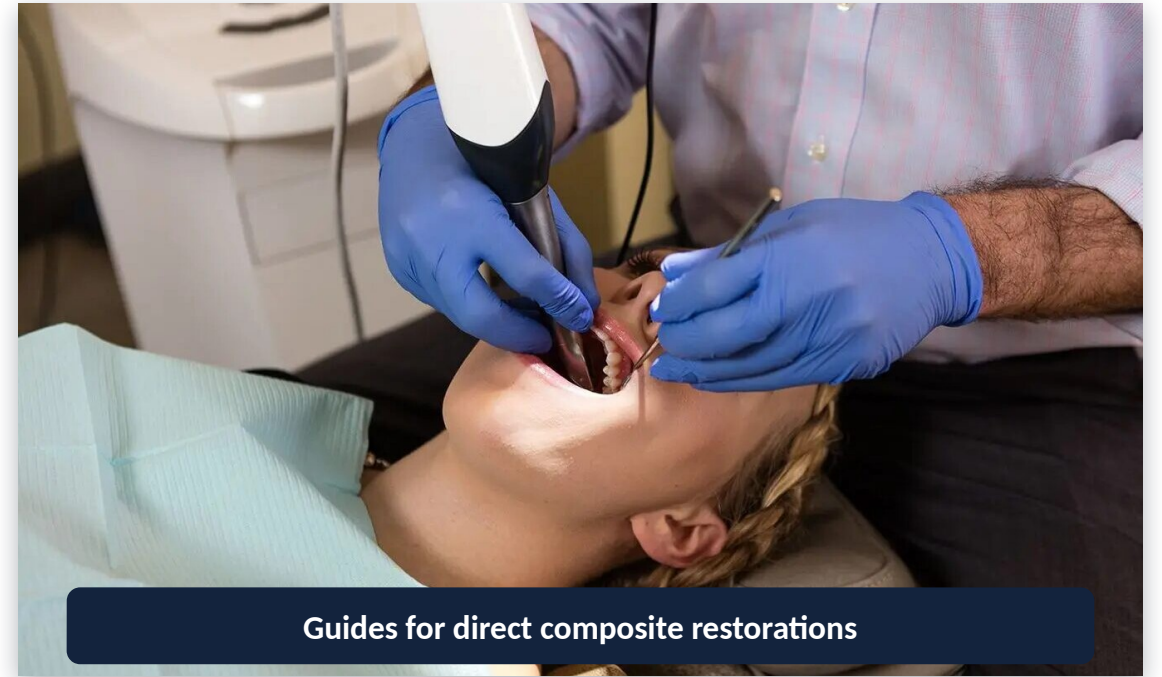
*For implants and direct composite restorations.*



# Two types of guides you can print



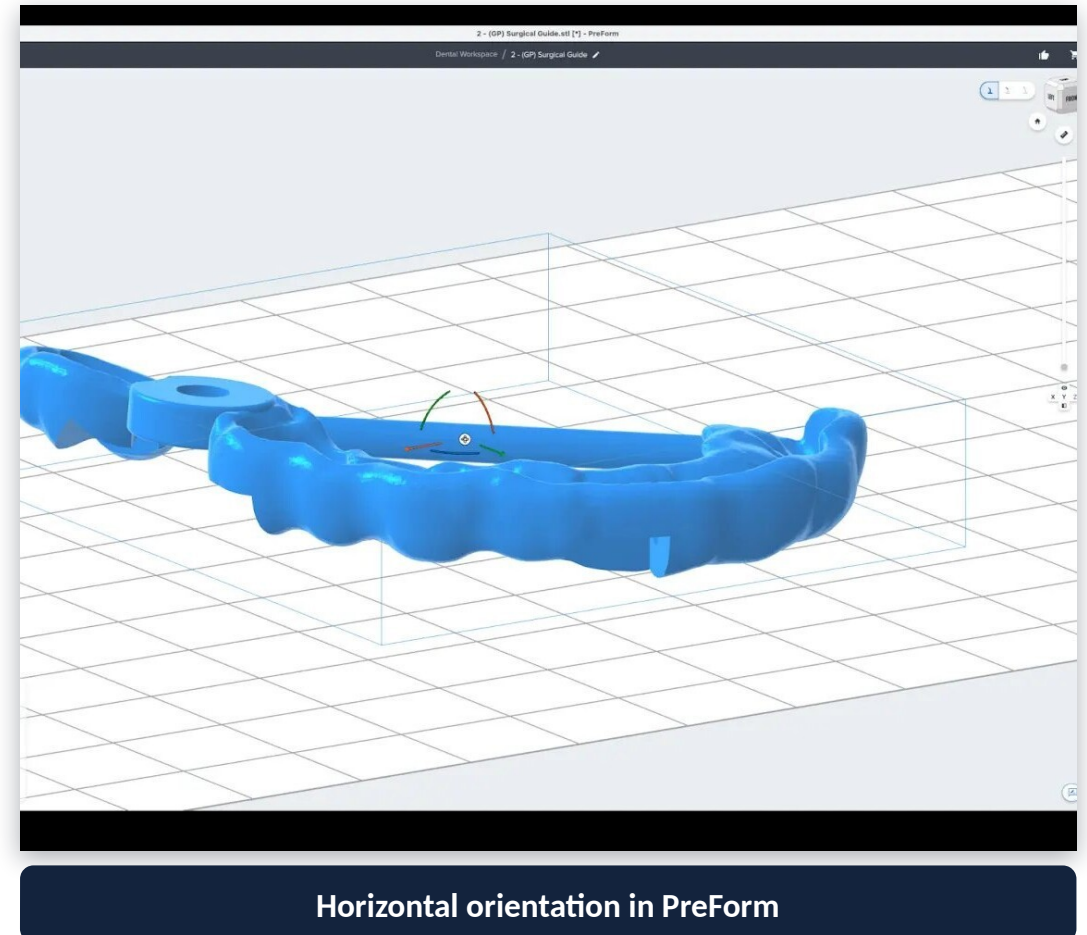
Planning based on intraoral scan + CBCT: implant position, angulation and depth guided with precision.



Digital mock-up (Digital Smile Design) transferred to the mouth: guides the direct application of composite onto the teeth.

# Surgical guides for implants: orientation and supports

- ✓ Keep the intaglio surface (the one that will contact the patient's dentition or tissue) facing away from the build platform
- ✓ Automatic supports, never near the sleeve holes or on the inner surface
- ✓ Multiple guides in the same print? Stagger them on the platform, without overlapping
- ✓ Always check sleeve hole alignment before printing





# Surgical guides for implants: from print to clinical use

- ✓ Wash in isopropyl alcohol and dry thoroughly (at least 30 minutes in air)
- ✓ Cure, then carefully remove supports near the sleeve holes
- ✓ Insert the sleeves: they friction-fit and stay in place on their own
- ✓ Sterilize in an autoclave: 132 °C / 4 min or 121 °C / 30 min



Removing supports near the sleeve holes



Finished guide

## ! Always respect the drying cycle after autoclaving

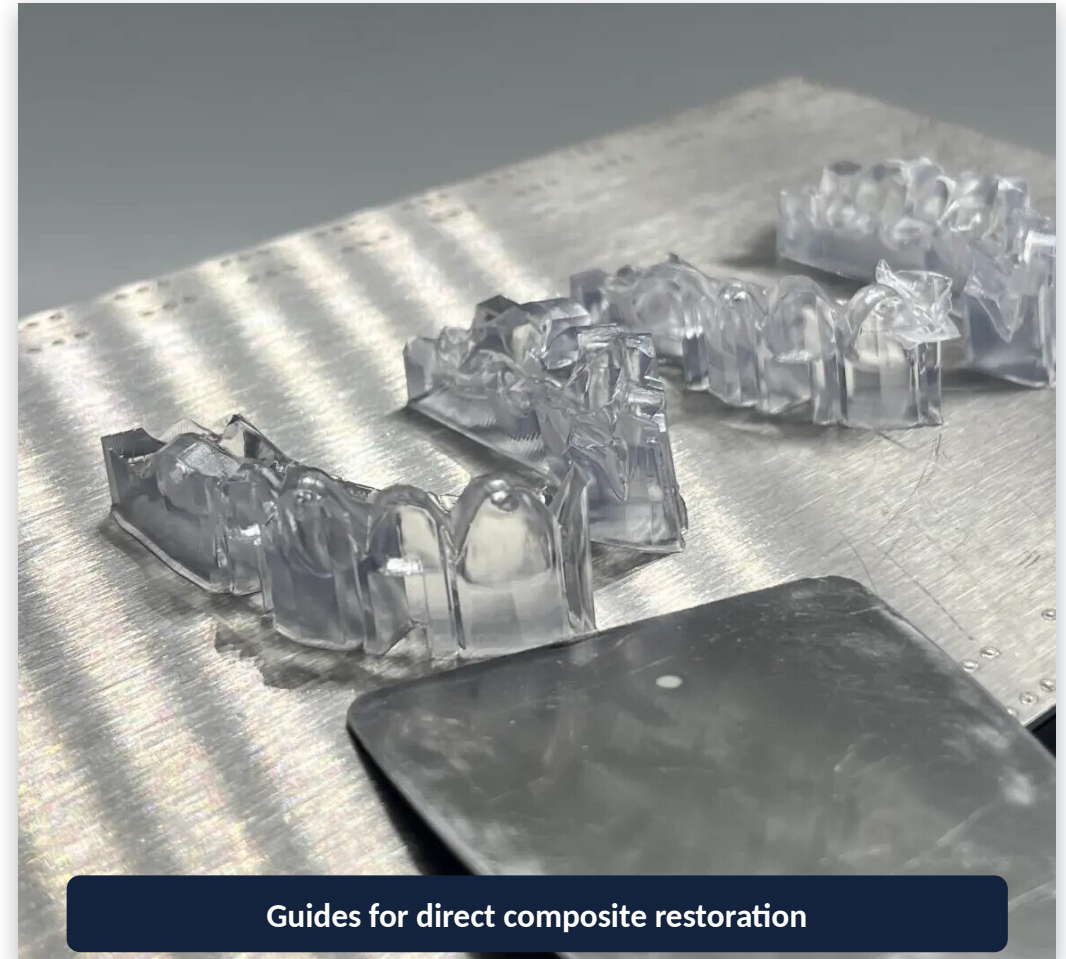
Autoclave cycles must include a drying cycle to ensure maximum precision. Instruments packaged and sterilized in a pre-vacuum autoclave should be dried for 20–30 minutes per the Centers for Disease Control (CDC) recommendations.

# Guides for direct composite injection

- ✓ These are a different product from surgical implant guides: they guide the direct application of composite (injected or pressed) according to the digital mock-up (Digital Smile Design)
- ✓ Orientation: intaglio surface facing outward, part parallel to the platform
- ✓ Printing directly on the platform? Set supports to "None" and disable internal supports

## Always cure underwater

Submerge the parts in a clear container full of water inside the curing unit — not dry, as with other resins.



Guides for direct composite restoration



SECTION 05

# 05

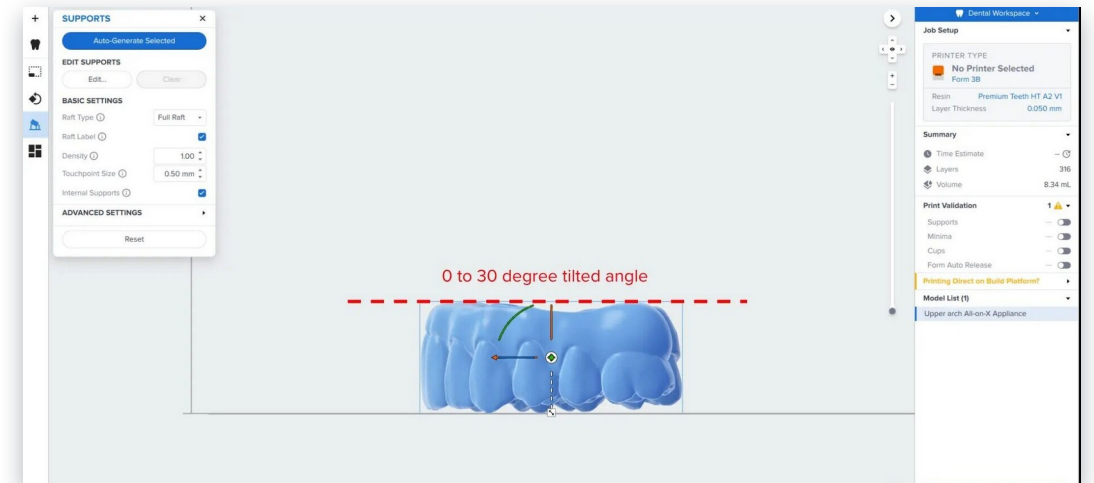
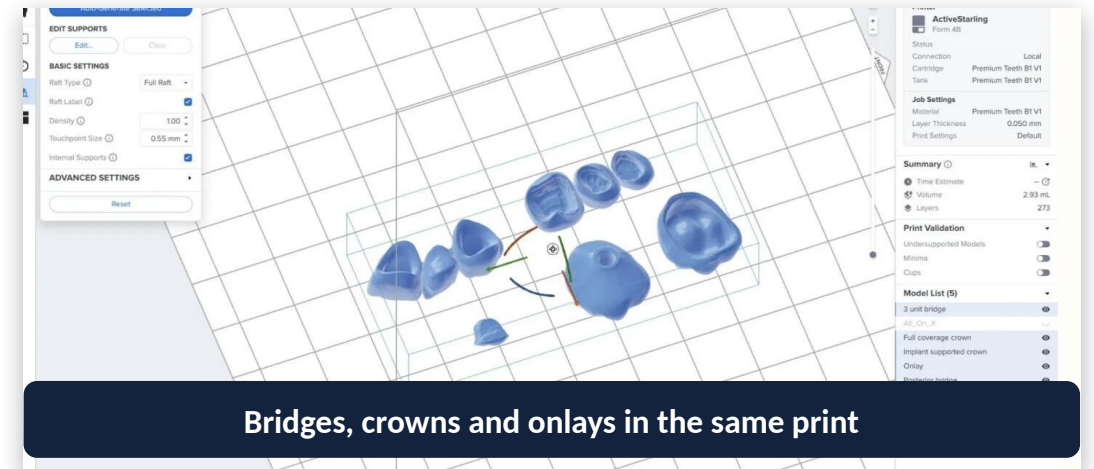
## Restorations: Provisional and Permanent

*Crowns, bridges and full arches — from orientation to delivery.*



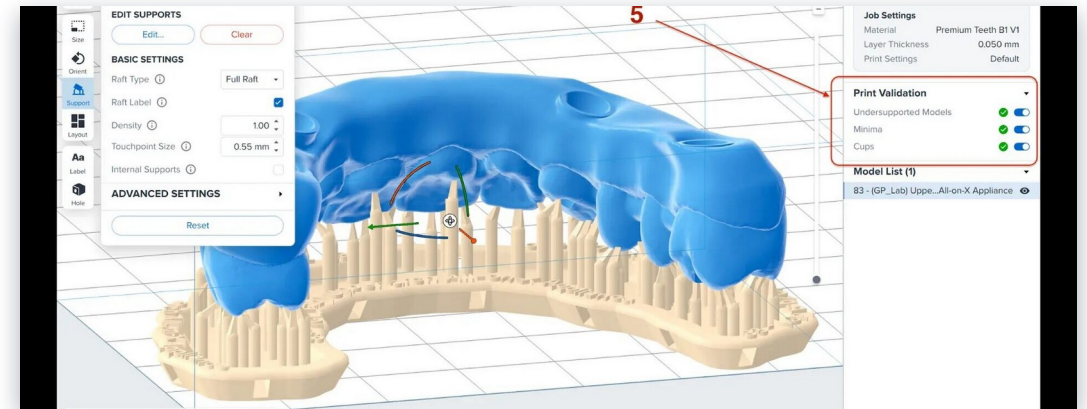
# Orientation in PreForm

- ✓ Occlusal surface always FACING the build platform
- ✓ Intaglio surface facing outward
- ✓ For anteriors: tilt slightly to protect the facial aesthetics
- ✓ Full arches (All-on-X): 0–30° angle, anterior teeth closer to the platform

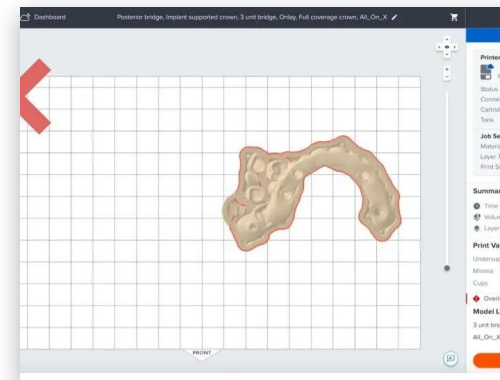


# Supports and layout: simple nesting

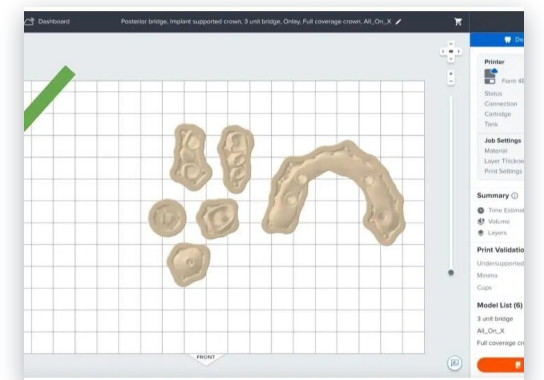
- ✓ Generate supports automatically, then manually refine the critical areas
- ✓ Move support points away from intaglio areas, contact points and connectors
- ✓ Inlays: place supports on the contact side with the adjacent tooth — it's still an area to refine manually
- ✓ Never overlap parts with each other — always check the green printability icon



Automatic support generation



✗ Overlapping



✓ Separated

## From printer to delivery (1/3)

1

### Remove

Detach the part from the platform with the dedicated tool or the release handles

2

### Wash

Isopropyl alcohol (IPA)  $\geq 99\%$ , parts fully immersed, in the Form Wash or ultrasonic cleaner

3

### Dry

At least 30 minutes in a well-ventilated area: check that no alcohol, uncured resin or residual particles remain

4

### Cure

With base and supports still intact, base side down, per the manufacturer's parameters

## From printer to delivery (2/3)



### 5 • Remove supports

Cutting disc or low-speed bur, removed one at a time — never tear them off: risk of chipping



### 6 • Polish

Traditional tools and pastes for acrylics and composites — never overheat the part



### 7 • Characterize

Pink composite for the gingiva, light-cure enamel/pigments for the tooth



# Minimum thicknesses: why they're not a minor detail

- ! Below the indicated minimum thickness, the risk of fracture during support removal, finishing or clinical use increases significantly
- ! Walls that are too thin deform more easily during washing and curing, compromising the restoration's precision
- ! The values on the next slides apply to single units, bridges and full implant-supported prostheses (All-on-X): always verify them in the design software before exporting the file

## Rule of thumb

Increase the thickness, don't reduce it: a thickness greater than the minimum rarely causes clinical problems, an insufficient thickness causes frequent fractures.

# Minimum thicknesses: single units and bridges

## SINGLE UNITS (CROWNS, INLAYS, ONLAYS, VENEERS)

|                        |        |
|------------------------|--------|
| Minimum wall thickness | 1.0 mm |
|------------------------|--------|

## BRIDGES (UP TO 7 UNITS)

|                                          |        |
|------------------------------------------|--------|
| Minimum wall thickness — anterior sector | 1.0 mm |
|------------------------------------------|--------|

|                                           |        |
|-------------------------------------------|--------|
| Minimum wall thickness — posterior sector | 1.5 mm |
|-------------------------------------------|--------|

|                                                        |                    |
|--------------------------------------------------------|--------------------|
| Minimum connector cross-section area — anterior sector | 12 mm <sup>2</sup> |
|--------------------------------------------------------|--------------------|

|                                                         |                    |
|---------------------------------------------------------|--------------------|
| Minimum connector cross-section area — posterior sector | 16 mm <sup>2</sup> |
|---------------------------------------------------------|--------------------|

|                     |         |
|---------------------|---------|
| Maximum bridge span | 7 units |
|---------------------|---------|

|                      |           |
|----------------------|-----------|
| Maximum pontic width | one molar |
|----------------------|-----------|

# Minimum thicknesses: full implant-supported prostheses (All-on-X)

FULL IMPLANT-SUPPORTED PROVISIONAL PROSTHESES

|                                                   |                                           |
|---------------------------------------------------|-------------------------------------------|
| Minimum wall thickness                            | 1.0 mm                                    |
| Minimum thickness around TiBase / custom abutment | 1 mm                                      |
| Cement space                                      | per the<br>implant manufacturer's library |



Always disable bur compensation, whichever design software you use.

# Design parameters: 3Shape

## CEMENT AND MARGIN SETTINGS

|                               |          |
|-------------------------------|----------|
| Cement space                  | 0.01 mm  |
| Extra cement space            | 0.07 mm  |
| Distance from the finish line | 1 mm     |
| Finish line width offset      | 0.15 mm  |
| Bur compensation              | disabled |

# Design parameters: exocad (1/2)

CEMENT AND MARGIN SETTINGS

|                      |         |
|----------------------|---------|
| Space                | 0.06 mm |
| From the finish line | 1 mm    |

ADDITIONAL SPACING

|        |         |
|--------|---------|
| Axial  | 0.01 mm |
| Radial | 0.01 mm |



# Design parameters: exocad (2/2)

EDGE

|                  |         |
|------------------|---------|
| Horizontal       | 0.15 mm |
| Angled           | 0 mm    |
| Angle            | 0°      |
| Vertical         | 0 mm    |
| Below the margin | 0 mm    |

ADVANCED

|                 |          |
|-----------------|----------|
| Pre-mill offset | disabled |
|-----------------|----------|

# The 5 things to always remember

*Valid for models, bite splints, surgical guides and restorations — the page to keep next to your software.*

1

## Models directly on the platform

anatomical surfaces always go on full bases, with supports for secure adhesion

2

## Functional surfaces without supports

3

## No overlapping between parts

always check the green printability icon before printing

4

## 50 µm layers for clinical precision

standard for models, bite splints, surgical guides and restorations

5

## Check the material's IFU and SDS

before every new clinical use: stated use, contact duration, absence of ACMO/TPO

# Questions?

*We're available for any further details.*

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# Documents and Guides Used

*This guide summarizes and simplifies the manufacturers' official instructions. For complete parameters, always refer to the original documents.*

- Formlabs Dental — Operating Manual: 3D Printing Dental Night Guards and Bite Splints
- Formlabs Dental — Operating Manual: 3D Printing Surgical Guides
- Formlabs Dental — Surgical Guides for Direct Composite Restorations
- Formlabs Dental — Operating Manual: 3D Printing Permanent Crowns (BEGO VarseoSmile TriniQ Resin)
- Formlabs Dental — 3D Printing Provisional Applications: Full Arches, Single Units and Bridges
- Asiga — "An Ultra production upgrade that changes the post-processing equation: UltraGLOSS / LIFT"
- Asiga Composer Quick Guide (internal document) — Composer 2.1 Handbook
- Regulation (EU) 2017/745 (MDR) on Medical Devices · ISO 10993-1:2018